Naman Tyagi

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Education

KIET Group of Institutions, Delhi NCR **Sept 2024 – Present**
Bachelor of Technology, Computer Science | ***75.13%*** *Ghaziabad, India*

DN Polytechnic, Meerut **2021 – 2024**
Diploma in Civil Engineering | ***67.48%*** *Meerut, India*

Technical Skills

Languages: C++, JavaScript, SQL, HTML, CSS
Frameworks/Libraries: React.js, Node.js, Express.js, Tailwind CSS, Prisma ORM
Databases: PostgreSQL (pgvector), MongoDB
Tools & Platforms: Git, GitHub, Redis, BullMQ, Docker, Postman, Vercel, Render
Core Concepts: Data Structures & Algorithms, OOP (C++), DBMS, Operating Systems

Projects

GitRAG | *AI-Powered GitHub PR Impact Analysis Tool* **Aug 2026 - Present**

- **Tech Stack:** Node.js, Express, PostgreSQL, pgvector, Prisma, Tree-sitter, BullMQ, Redis, GitHub API/OAuth
- Architected a **full-stack SaaS tool** that analyzes GitHub pull requests using **AST parsing** and **dependency-graph traversal** to predict downstream code breakage before merge, automatically posting results as PR comments via the GitHub API.
- Designed a **hybrid retrieval system** combining **PostgreSQL full-text search (BM25)** with **pgvector cosine similarity**, tuning confidence thresholds against real usage data to separate genuine semantic matches from same-codebase noise.
- Implemented AST-based code intelligence using **Tree-sitter**, parsing JavaScript/TypeScript into structured symbols and building a call-dependency graph with **recursive CTE traversal in PostgreSQL** to identify direct and transitive callers up to N hops deep.
- Built an **async job pipeline** with **BullMQ and Redis** to process GitHub webhook events, decoupling webhook response time from multi-stage analysis (ingestion, embedding generation, impact analysis).
- Implemented **GitHub OAuth 2.0** and a self-serve repo-connection flow, enabling users to programmatically register webhooks on their own repositories with real-time ingestion status tracking.

AlgoArena | Live Link **May 2026 - June 2026**

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Stream.io SDK, Wandbox API
- Architected a **full-stack collaborative interview platform** with **real-time video**, chat, and **role-based access control** for students and administrators.
- Engineered a **dual-layer code execution pipeline** with visible test-case feedback for candidates and **hidden server-side test evaluation**, preventing test-case leakage and ensuring fair assessment.
- Secured all **Express API endpoints** using **JWTs in httpOnly cookies**, bcrypt password hashing, and custom **RBAC middleware**.
- Built a resilient **real-time communication layer** using **Stream.io**.

Threadly | Live Link **2025**

- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Stripe, JWT, Cloudinary
- Developed a full-stack e-commerce application with **Role-Based Access Control (RBAC)** for admin and user modules.
- Built a **real-time cart system** using React Context API with server-synced persistence in MongoDB.
- Implemented **multi-level category/subcategory filtering**, price-based sorting, and live search for a responsive shopping experience.
- Integrated **Stripe payment gateway** with server-side Checkout Sessions and a dedicated Admin Panel for inventory/order management via Cloudinary-backed image uploads.

Achievements

Problem Solving (LeetCode & GeeksforGeeks) **Ongoing**

- Solved **450+ Data Structures and Algorithms problems** across LeetCode and GFG; achieved a LeetCode **Max Rating of 1654**.

Smart India Hackathon (SIH) 2024 – Departmental Winner **Sept 2024**

- Selected in the departmental round among 120+ teams as **Backend Developer** for PMSSS, a scholarship disbursement system; implemented **RBAC and JWT-based authentication middleware**, **OCR-based document verification**, and deployed the application on Render.